

Reviewing domain-based approaches for misinformation research

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To study the spread of misinformation on digital platforms, it is imperative for computational social scientists to have access to reliable and valid tools that enable them to identify misinformation. A prominent way to track misinformation is to use a list of domains. For example, the organization NewsGuard offers the most comprehensive database of domain trustworthiness ratings and is increasing in popularity. Despite restrictive access, the database was never created as a research instrument and has never been reviewed systematically. Therefore, we offer a comprehensive description and inspection of the NewsGuard database, across the following dimensions: the reliability of trustworthiness ratings, the selection of domains, as well as the completeness of categories such as political orientation. By describing the database, we outline potential pitfalls and recommendations for digital media research and discuss appropriate use cases.

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Introduction

Misinformation permeates both social media platforms (Lazer et al., 2018) and mainstream media channels (Tsfati et al., 2020). However, only a fraction of news consumption is attributed to false news — roughly 0.15% (Allen et al., 2020) — and only a small group of users actively engage with unreliable information online (Grinberg et al., 2019). Typically, these individuals hold strong political or ideological beliefs, accepting information that echoes their opinions and often coming from established sources. Consequently, it is unsurprising that misinformation exacerbates partisan divides and fuels inter-group hostility (González-Bailón et al., 2023; Robertson et al., 2023), capitalizing on these dynamics to stoke conflict and evoke negative emotions. With decreasing levels of trust in the media (Newman et al., 2020) and institutions (Bennett & Livingston, 2020), it has become more critical to understand such dynamics reinforcing the dissemination of misinformation and their relationship with digital media.

The spread of misinformation has been studied extensively by scholars from various scientific disciplines, relying on diverse conceptualizations. In psychology, researchers commonly define misinformation as inaccurate or unreliable information (Ecker et al., 2022), while in communication science, a popular definition is that by Claire Wardle and Hossein Derakhshan (2017), distinguishing misinformation from disinformation and fake news that pursues financial or political intentions. Computational social scientists, for instance David Lazer and colleagues (2018), follow this notion: "Fake news overlaps with other information disorders, such as misinformation (false or misleading information) and disinformation (false information that is purposely spread to deceive people)." (Lazer et al., 2018, p. 1094) Inconsistencies in definitions leave researchers with certain degrees of freedom, mostly concerning the boundaries of misinformation, or where to draw the line on truth. Conceptual choices, in reality, largely depend on methodological constraints or preferences by journals or conferences, which is why the terms are often used interchangeably. This has led to imprecise or inconclusive findings around the prevalence of misinformation and its effects. While Altay et al. (2023) and many others make the observation that misinformation is overstated, Tay et al. (2024) argues that heterogeneity in how we measure misinformation results in heterogeneity in evidence and creates a false dichotomy of understanding misinformation as a symptom or a cause.

To study the diffusion or discussion of misinformation on social media, computational researchers commonly follow a selection of stories or sources, thereby investigating the phenomenon at different levels of analysis. Both approaches typically rely on experts, such as fact-checkers, to curate a list that is then used as a starting point to collect data from one or more digital platforms. The choice between story- and source-based approaches hinges on the primary research objective: When operating at the level of the story, there is low variance in accuracy ratings and we can capture misinformation with high precision, i.e., we can be sure that what we measure is related to false information. However, the selection of stories can be narrow, and it is nearly impossible to generalize findings beyond that specific, potentially biased case. For instance, Vosoughi et al. (2018) selected a few fact-checked fake news stories. This has the advantage that the researchers can rely on experts to determine what is objectively false; the researcher does not have to judge the trustworthiness of the stories themselves. On the other hand, misinformation taken from fact-checking websites is common and often widely debated and covers topics that fact-checking organizations have experts for in this specific moment. Therefore, the sample only represents a small proportion of misinformation and neglects more subtle types of misinformation that are harder to identify. While the story-level approach captures dynamics related to some types of misinformation, results tend to underestimate both the reach of misinformation and the effects thereof.

Conversely, source-based strategies can cover a wider variety of sources and stories, and hence, samples scale up to a more diverse information environment. By sampling based on sources, we can include more subtle types of misinformation and get accurate estimates both for the scale (Allen et al., 2020; Grinberg et al., 2019; Guess et al., 2020) and dynamics of its spread (Lasser et al., 2022; Robertson et al., 2023) add: Pennycook et al., 2021. But even if we overgeneralize across articles and error rates for article accuracy are high, source judgments can still be unbiased at the macroscale. The precision of such approaches depends on the coverage, but research has shown that ratings for U.S.-based sources overlap (Lin et al., 2023). Now, being conceptually rigorous, misinformation here is operationalized as untrustworthy sources, and observed patterns may be less related to misinformation and rather to the source, their political bias or editorial practices. A lack of editorial practices may lead to lower quality in news stories (Lazer et al., 2018). Unreliable publishers publish mostly unreliable news pieces, but they may intentionally post reliable news pieces once in a

while to attract new readers. In order to exclude other source-related factors, this approach can take source biases and misleadingness into consideration while also being generalizable. In reality, however, categorizations of sources are made on a binary scale, neglecting more subtle forms of misinformation. Furthermore, this argument ignores that source ratings are sensitive to time and context. Sources may change, for instance, due to ownership. In addition, while source judgments can be made irrespective of topic expertise, such ratings are often based on journalistic norms of one media system that are not necessarily applicable to other contexts.

In conclusion, we see source-based approaches as a promising way to track misinformation online. However, for computational social scientists to study the spread of unreliable news, they require access to reliable and credible databases of source ratings. Since the 2016 U.S. elections, the most prominent way to track misinformation online has been to use a list of untrustworthy or unreliable domains (e.g., Grinberg et al., 2019). The organization NewsGuard offers the most comprehensive database, not just blacklisting domains but providing a trustworthiness score per domain. A team of journalists and editors rate the domains based on nine journalistic quality criteria and regularly updates both the coverage of the list, i.e., the number of domains, as well as their overall trustworthiness rating. Recently, the database has been extended to other countries than the United States and is constantly growing. Recently, the database has increased in popularity. For instance, it has been used to study user's engagement with unreliable news in the context of the 2018 and 2020 US elections Robertson et al. (2023) or to assess if members of the Republican party in the U.S. Congress have contributed to spread of misinformation by linking to untrustworthy sources Lasser et al. (2022). However, the database was created as a tool for brand security rather than research, which is why access is paywalled and expensive. In the light of the discussed disadvantages of list-based approaches, we here provide a comprehensive description of the content of this database following the research question: How reliable is the NewsGuard database of domain trustworthiness ratings over time and language contexts?

More precisely, we believe that it is essential to address the following aspects of the NewsGuard database in more detail: First, we look into the distribution and update process of the trustworthiness as well as the selection and removal of sources over time. Second, we assess the reliability of the database for different countries, and lastly, we estimate

the usefulness of other information provided in the database, e.g., journalistic criteria and political orientation. Therefore, we focus on three questions:

How volatile are trustworthiness ratings, e.g., how often do ratings change or do domains vanish? First, we investigate the development of the NewsGuard database over time. This is important to estimate whether using an older database will distort results. We describe the different versions of the database and reproduce published research relying on the NewsGuard list with different versions of the list to assess the degree to which results depend on a specific version of the list.

How reliable and complete is the provided information, particularly for non-US countries? A recent study has shown high agreement of NewsGuard with other expert-rated lists in the U.S. context (Lin et al., 2023), but an assessment for other countries and contexts is missing. Therefore, we next compare the distribution and completeness of information provided by NewsGuard across different contexts.

How can other information be used for misinformation research? For each source, NewsGuard experts rate journalistic quality criteria, topics, and political orientation, but they also label AI-generated content. Estimating to what extent the different criteria constitute the overall trustworthiness score and the completeness of additional information might shed light on aspects of misinformation beyond binary true/false labels.

By describing the database, we outline potential pitfalls for digital media research and offer recommendations for the application of the NewsGuard database in different contexts. We also want to empower misinformation researchers to make informed decisions about whether the substantial expense associated with acquiring access to NewsGuard is necessary for their research.

Dataset construction

As of November 15, 2023, the database contains 11,093 source assessments. Some sources, however, do not receive a trustworthiness score as they are classified as a “platform” (63 domains, e.g., YouTube). In addition, the database includes 63 sources judged as “satire” (e.g., The Onion) and 454 lifestyle sources (e.g., fitworldnow.com). NewsGuard

does not rate these sources, but they make up less than 1% of the listed sources in the database. We exclude them from further analysis. For the other 10,513 rated sources, the database includes the following types of ratings (see Table SX for a full list of variables):

- Domain and trustworthiness rating (between 0 and 100).
- Binary labels for nine journalistic quality criteria constituting the trustworthiness rating.
- Language and country associated with the domain.
- Political orientation of the outlet.
- Topics covered (e.g., “local news”, “political news”, “health information”) by the outlet
- A flag indicating whether the outlet frequently posts AI-generated content.

A trained team of NewsGuard experts regularly judges the trustworthiness of the selected news sources. The organization also manually adds sources they deem influential or that changed domains. The database automatically updates every hour and allows us to track changes going back to March 2019, when the organization started operating. On their website, NewsGuard claims to have reviewed domains accounting for 95% of online engagement. Still, the procedure by which domains are selected or removed is undisclosed on the website, as are the intervals in which the trustworthiness assessments are updated in the database.

Descriptive Analyses

How volatile are trustworthiness ratings?

In the following, we review how the database, and particularly the trustworthiness ratings and domain selection, have changed over the years. **Coverage.** Over the span of four years, NewsGuard has grown from 2,615 to 10,513 rated sources, as shown in Figure 1. We sampled the database in monthly intervals and chose the middle of the month for the snapshot. However, when we sampled from the second day of each month, we observed

an irregular dip in the number of domains (see May 2022 in Figure 1). We think an error caused this fluctuation, as the approximately 3,500 removed domains were added again two weeks later (see Appendix). In our case, changing the sampling procedure to mid-month shows to be more stable. Nevertheless, the substantive loss in the number of domains caused by one difference in the sampling strategy indicates that comparing multiple snapshots of the database can be decisive.

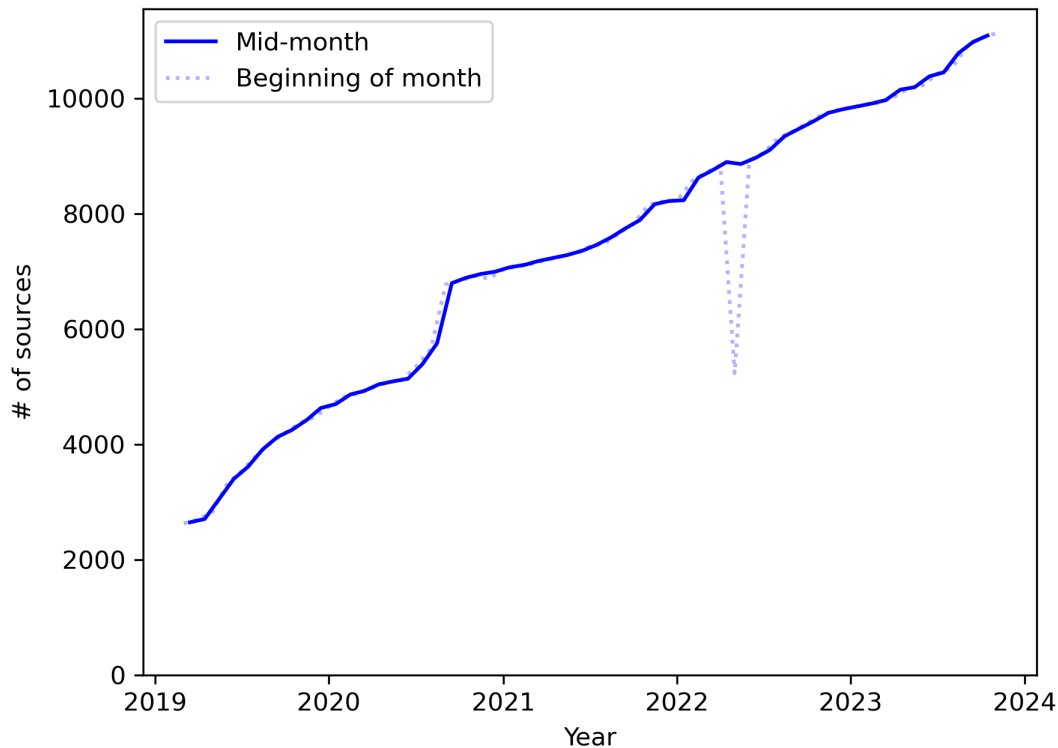


Figure 1. Number of unique domains in the NewsGuard database over time

Distribution. NewsGuard trustworthiness ratings are assigned on a scale from 0 to 100, with 100 being trustworthy. The threshold for the trustworthiness of a source is set at 60. Notably, NewsGuard is not a blacklist. In fact, close to 60% of domains have an overall rating of over 60 and are therefore considered “trustworthy”(see Fig. 2) for the distribution of trustworthiness scores.

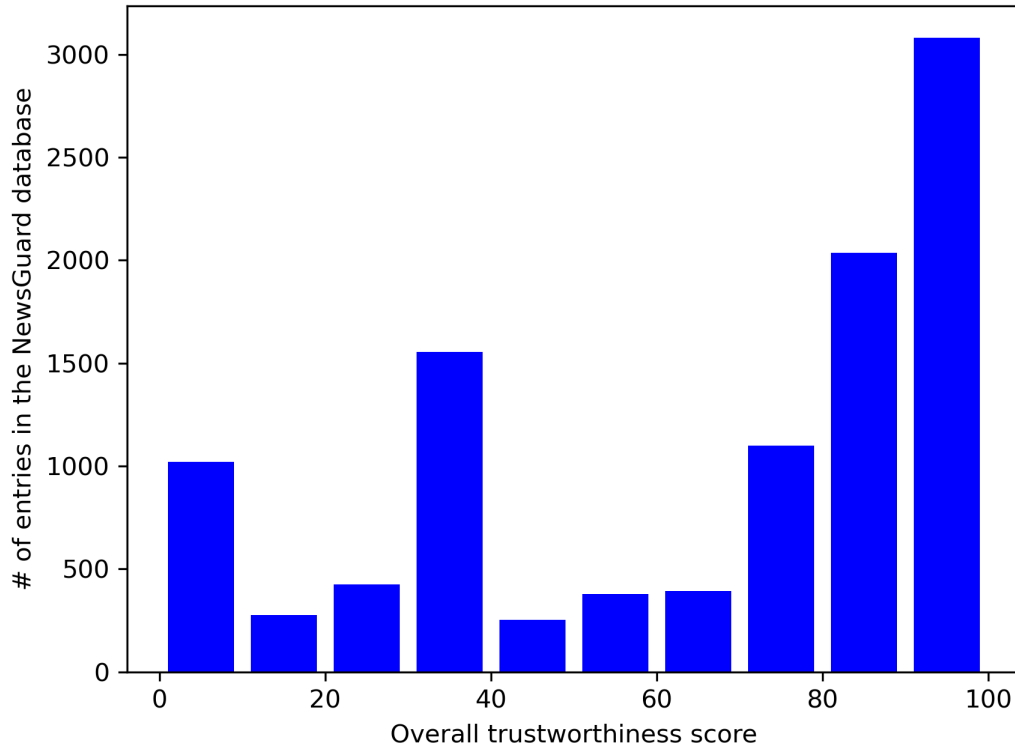


Figure 2. Distribution of domain trustworthiness contained in the NewsGuard database on November 15, 2023.

When looking at the score distribution over time, the left skew has been diminishing: while the average score was 70.79 (SD=35.33) in 2019, it is now at 64.61 (SD=32.33). In other words, the average trustworthiness of domains has decreased over time. The distributions of trustworthiness from 2019 to 2023 are shown in Figure 3.

NewsGuard regularly adds new sources to the database but rarely removes any, i.e., the database has been growing to three times its initial size. When comparing the added domains for one month to the previous month, NewsGuard has added 7,705 sources over time while they only removed 612. Removed sources tend to have a slightly lower trustworthiness score (M=53.50, SD=31.60) than added sources (M=62.09, SD=32.14). A great number of sources with lower ratings (M=46.5, SD=31.62) were added in 2020, which

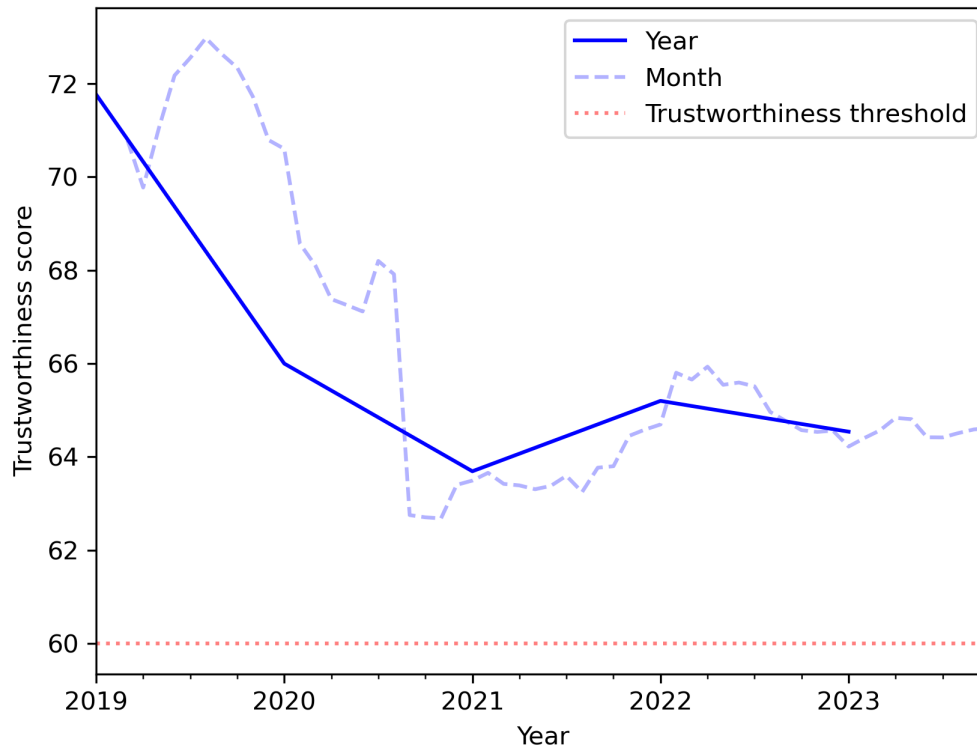


Figure 3. Distribution of trustworthiness scores over time

might explain the drop in trustworthiness in the following.

Each entry in the database has a unique identifier and NewsGuard indicates the date when a rating was last updated, even if the score did not change. Updates happen on average once a year (279 days), but this varies strongly per domain, with a maximum of 1,109 days. When updated, trustworthiness scores mostly remain untouched, i.e., most sources have not decreased in trustworthiness. However, some domains drastically dropped in trustworthiness; for instance, *conservativedailynews.com* lost 80 points in trustworthiness in one update. Figure 4 illustrates the differences in consecutive scores over time, with major score updates in early 2020.

Reproducibility. When reproducing results of XYZ...

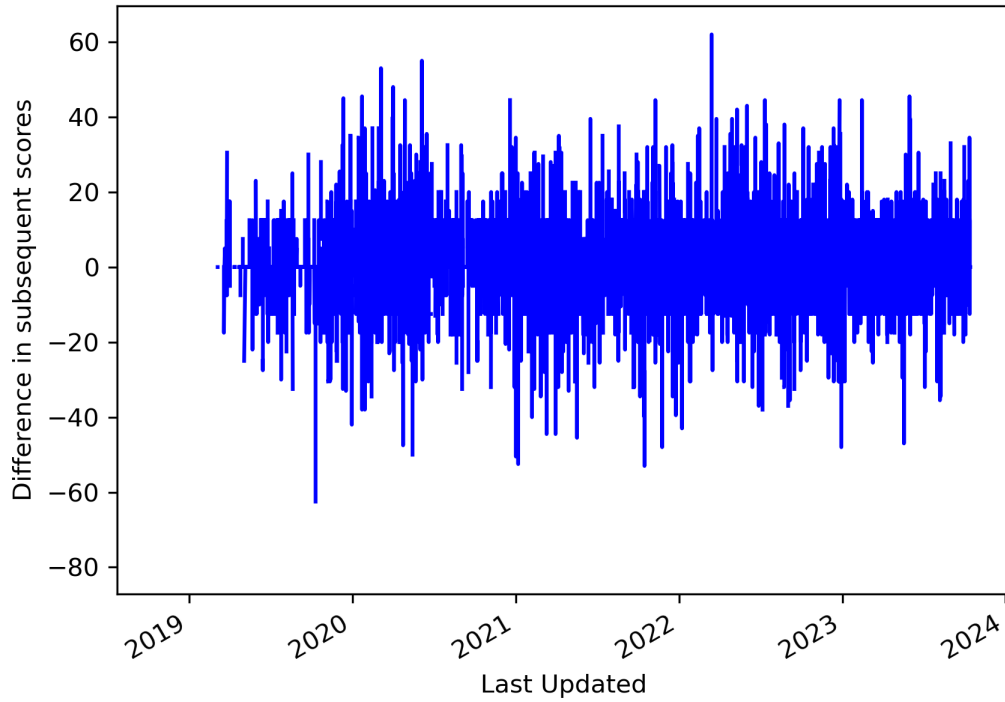


Figure 4. Changes in trustworthiness scores over time, calculated as the difference in scores per domain between consecutive updates.

How reliable and complete is the provided information, particularly for non-US countries?

Country coverage. The dataset contains sources from nine countries in total (United States, Great Britain, Italy, Canada, France, Germany, Austria, Australia, and New Zealand) plus sources considered to be global. The majority of sources are from the United States (75.04%), followed by global sources (14.05%), Great Britain (6.4%), Italy (5.28%), Canada (4.91%), France (3.8%) and Germany (3.59%), and Australia (1.3%). The remaining countries, Austria and New Zealand, account for less than 1% of sources each. NewsGuard also clusters sources based on language (English, Italian, French, German, Spanish). See Table ?? for an overview of the number of sources per country and language.

Trustworthiness distribution. Across all versions of the database and most countries covered by NewsGuard, the overall trustworthiness of sources remains >60 , except for U.S. sources: U.S.-based outlets are of lower quality on average. Figure 5 shows the average trustworthiness of sources per country over time. Another reason for the lower trustworthiness score could be related to the relative number of outlets in the dataset. In other words, the U.S. coverage seems the most complete and reliable, given the total number of outlets rated by NewsGuard as well as the frequency of updates. While the U.S. database was updated 32,062 times in total (5,346 updates per year on average), the maximum number of times that the second most comprehensive database, Italy, was updated is only 1,966.

How can other information be used for misinformation research?

Journalistic criteria. The trustworthiness score is a composite score that is calculated based on nine journalistic criteria. For each domain, the organization assigns a binary label (yes/no) per criterion. The trustworthiness score is then calculated as the sum of the weighted criteria. The criteria are weighted differently, ordered in descending relevance:

- “Does not repeatedly publish false content” (22 points)
- “Gathers and presents information responsibly” (18 points)
- “Regularly corrects or clarifies errors” (12.5 points)
- “Handles the difference between news and opinion responsibly” (12.5 points)
- “Avoids deceptive headlines” (10 points)
- “Website discloses ownership and financing” (7.5 points)
- “Clearly labels advertising” (7.5 points)
- “Reveals who’s in charge, including possible conflicts of interest” (5 points)
- “The site provides the names of content creators, along with either contact or biographical information” (5 points)

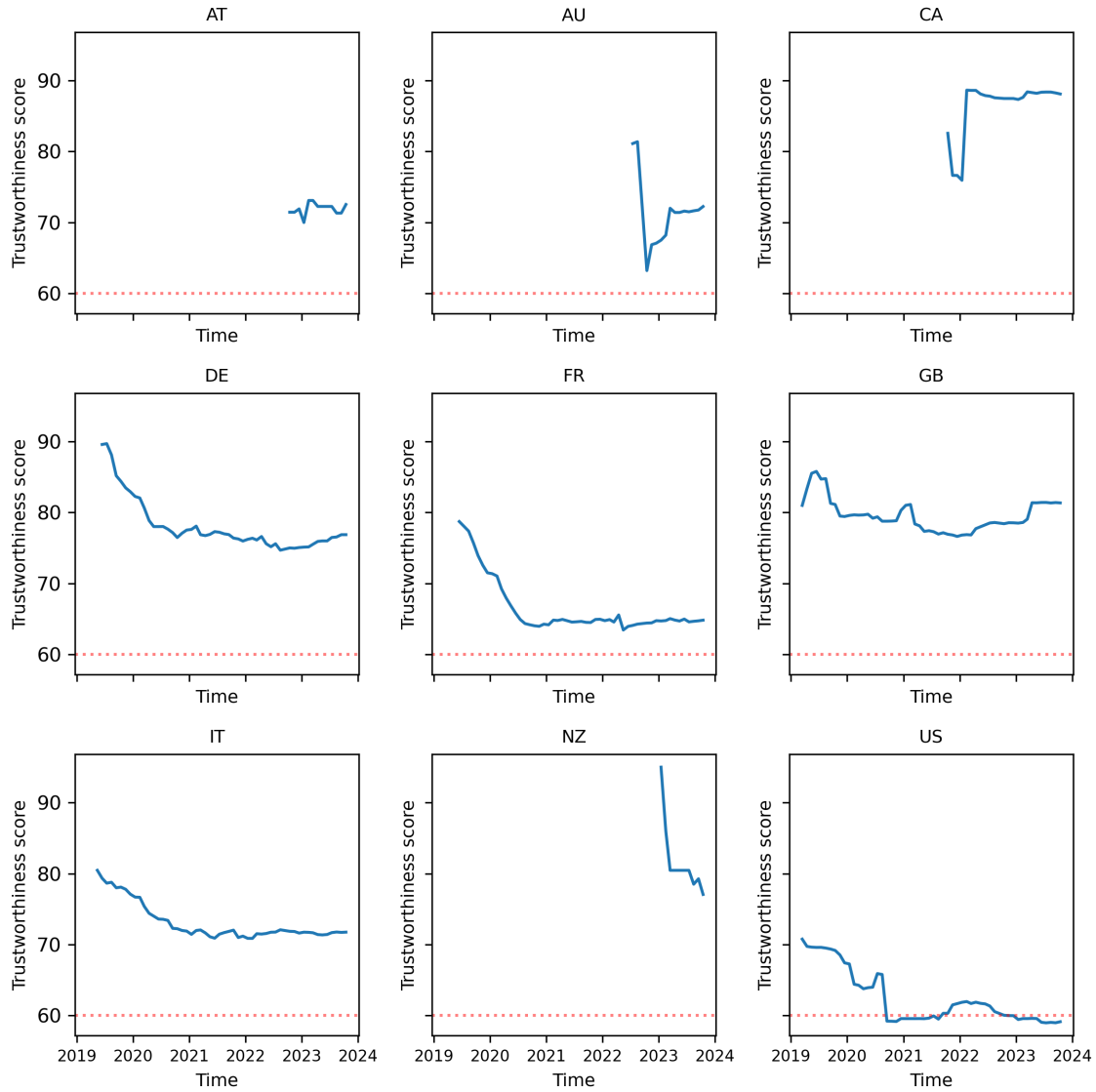


Figure 5. Monthly aggregates of trustworthiness scores over time, per country covered by NewsGuard.

Further explanation of the criteria and their weights can be found in the NewsGuard FAQ¹.

¹<https://www.newsguardtech.com/ratings/rating-process-criteria/>

In the latest version of the dataset, the criterion “Avoids deceptive headlines” was the most respected, and 8,597 of the sources satisfied it (78.9 % of sources), followed by “Does not repeatedly publish false content” (77.5 % of sources). Across all dataset versions, when one of the two criteria is satisfied, the other one tends to as well (Pearson’s $r = 0.88$). A factor analysis with oblique rotation confirms that the two criteria make up one factor (accounting for 56% of variance explained and with factor loadings > 0.9), while all other criteria share 35.9% of the variance explained (factor loadings $> .6$). Meeting one of two criteria strongly predicts a high rating, and vice versa (with coefficients of 64.74, $< .001$, and 67.28, $p < .001$, respectively). In line with that, 11.97% of the ratings only meet these two criteria; the most common combination of criteria, however, is that all criteria are satisfied (17.97% of ratings)—given that the majority of sources are considered trustworthy.

Furthermore, looking at domain criteria updates, we find that most sources also keep their scores. The most common update pattern is a negative change in the criterion “Website discloses ownership and financing”, occurring, for example, once for the German weekly newspaper zeit.de. This criterion appears to be the most volatile, with 13.26% of sources eventually changing their label negatively. On the other hand, a positive update is also the third most common pattern (with 10.77% of sources), indicating that criteria related to ownership might be a criterion that is frequently updated/monitored by NewsGuard.

Political Orientation. The dataset also classifies sources according to their political orientation. The classification relies on a left-to-right conception of the political spectrum and should be interpreted in the context of the respective political landscape. In December 2022, the labels shifted from four (far left/ slightly left/slightly right/far right) to only two possible labels (left/right).

Overall, the political orientation dimension covers only a minority of the sources in the database. Out of 9,811, 3,343 sources have a political orientation rating. The ratio decreases when zooming in on singular countries: only 42.87% of news outlets have a political orientation rating in the United States, 14.57% in Great Britain, 16.49% in Italy, 19.74% in Germany, 4.92% in Canada, and 10% in New Zealand. Here, again the United States is overrepresented in the dataset.

Trustworthiness ratings are lower for right-leaning sources. The mean trustworthiness score for right-leaning sources is 58.5, while it is 66.5 for left-leaning sources. The median is 65 for right-leaning sources and 77.5 for left-leaning sources. The distribution of scores is shown in Figure X.

Topics. The database assigns topic labels, which are based on the topics covered by the outlets. The topics are not mutually exclusive, i.e., a source can be assigned to multiple topics. The topics are, for instance, “Local News”, “Political News or Commentary”, or “Conspiracy Theories or Hoaxes”. The topics are also not exhaustive, as 1,366 sources are not classified under any topic (see Table for topic occurrence).

Trustworthiness ratings greatly differ across topics. Figure X shows that sources covering conspiracies have the lowest average rating, while those covering “Technology” (M=87.16) and “Business or Personal Finance” (M=83.21) have the highest average ratings. The category “Political News and Commentary” attracts the highest share of bad sources: 80.09% of all untrustworthy sources are said to cover that topic, making up the 76% of all sources covering that topic.

The “Conspiracy” label contains only 39% of the untrustworthy sources, suggesting that it only makes up a small proportion of all untrustworthy information. Nonetheless, it is a highly typical topic for bad sources: 99% of the sources connected to this topic are untrustworthy. The quality of sources linked to each topic is illustrated in Figure X.

AI-labelled content.

Conclusion

Points: Right-skewed distribution of scores reflects the presence of misinformation. Even if there are more sources out there, their prevalence should be small. Misses user-created misinformation. Monthly snapshot of database should be enough to cover changes.

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Online Appendix

Appendix goes here.